25900054 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCFS102 - INTRODUCTION TO COMPUTER NETWORKS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Describe the role of ICMP in the TCP/IP model. [U] / [CO1]
2. Identify the network topology that is easiest to troubleshoot. [An] / [CO1, CO2]
3. Explain the role of a router in a WAN. [U] / [CO1]
4. Compare IPv4 and IPv6 headers. [U] / [CO1]
5. Differentiate between a primary server and a secondary server in DNS. [U] / [CO2]
6. Explain the purpose of the DATA command in SMTP. [U] / [CO2]
7. Simplify the concept of backpressure in congestion control. [An] / [CO2]
8. Identify the goal of a security services [U] / [CO1]
9. Identify the type of OSI security Architecture [U] / [CO1]
10. Identify the two type of access control [U] / [CO1]
11. Discuss deadlock in an OS [U] / [CO4]

12. Illustrate how volatile and non-volatile digital evidence differ in analysis [An] / [CO4]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

13. Identify the significance of network protocols in data communication [U] / [CO1]

14. Combine the sender's and receiver's actions in a checksum operation to explain how errors are detected. [An] / [CO2]

15. Describe the process of how hamming code can correct single bit errors. [U] / [CO2]

16. Explain Active and Passive Security attacks [U] / [CO1]

[5x3 = 15]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

17. Explain the different types of connecting devices [U] / [CO1]

18. Explain piggybacking in Go-Back-N ARQ. [U] / [CO2]

[10x1 = 10]